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FACTORIZATION AND SURPRISE AS LABEL-FREE CONTROL SIGNALS FOR JOINT-EMBEDDING PREDICTIVE ARCHITECTURES

Abstract. Joint-embedding predictive architectures learn by predicting latent targets, yet their training loss can improve while semantic quality stagnates or the representation collapses. We study two possible label-free control signals: *surprise*, for deciding which observations deserve updates, and *factorization*, for diagnosing whether relation-specific structure emerges in the latent space. The work is a sequence of hypotheses, controlled failures, metric audits, and progressively stronger experiments on a vector world, Shapes3D, Tiny ImageNet, and Moving-MNIST. Richness-aligned surprise rejects corrupted data, but concentrates sampling support and does not outperform uniform sampling; even a moderate-support variant lowers the JEPA loss while reducing Top-1 accuracy. Factorization is detectable: a three-dimensional Shapes3D subspace retains nearly all shape accuracy while suppressing context, and official I-JEPA checkpoints contain reproducible joint block structure beyond a spectrum-preserving Haar null. However, the structure is strongest early and decreases while downstream probes improve. LDA-, crop-, and operator-based scores correlate strongly with quality within one trajectory, but do not reliably select between training methods or forecast later quality. These results reject a universal scalar “representation quality” interpretation. We instead propose a relation-conditioned profile—capacity, predictability, modularity, support, and utility—and a causal program combining surprise with early, decaying factorization regularization under explicit collapse and sampling-support guardrails.

Key words and phrases: joint-embedding predictive architectures and self-supervised learning and representation factorization and surprise sampling and representation diagnostics.

§1. Introduction

Joint-embedding predictive architectures (JEPAs) predict the representation of a masked image region or future video region from visible context instead of reconstructing pixels. I-JEPA demonstrated that this objective can produce semantic image representations without handcrafted view invariances [1]; V-JEPA extended the principle to video and obtained useful frozen motion and appearance features [2]. The objective is attractive precisely because it may ignore unpredictable pixel detail. The same indirection creates an evaluation problem: a low latent prediction loss is neither a guarantee of semantic usefulness nor a complete warning against dimensional collapse.

The project asked whether the stop-gradient/EMA system has meaningful fixed-point dynamics, whether unlabeled “richness” can compare representations, whether surprise can allocate useful updates, and whether these signals can improve video pretraining. We treated each claim as a hypothesis and retained negative results and metric corrections. Two directions survived: *surprise*, a local state-dependent update signal, and *factorization*, relation-conditioned latent geometry.

We formalize and test both directions from synthetic mixtures to image and video JEPAs; audit factorization with held-out operators, fixed-dimension whitening, split-half projectors, and spectrum-preserving nulls; show that high same-run correlations do not imply cross-method selection; and propose causal, phase-specific experiments judged by downstream utility and checkpoint regret.

§2. Related Work

I-JEPA predicts masked target-block embeddings with an EMA target encoder [1]; V-JEPA extends feature prediction to spatio-temporal masks [2]. Deep-linear analysis suggests a bias toward high-influence predictable features [6]. For unlabeled diagnosis, RankMe measures marginal capacity [4], LiDAR adds relation-discriminative LDA structure [9], and VICReg explicitly guards variance and redundancy [3]. For data selection, RHO-LOSS distinguishes hard learnable points from noise and redundancy [7].

We connect these strands by testing a diagnostic within a run, across training rules, and as an intervention target.

§3. Problem Formulation and Hypotheses

Let f_U be the online encoder, f_C the stop-gradient target encoder, and g_U the predictor. For context x_i^c and target x_i^t ,

$$\ell_i(U, C) = \|g_U(f_U(x_i^c)) - \text{sg}(f_C(x_i^t))\|_p, \quad C^+ = mC + (1-m)PU, \quad (1)$$

with $p \in \{1, 2\}$ and parameter correspondence P . Raw surprise is $s_i = \ell_i$. Because large loss mixes learnable novelty with irreducible noise, we also use positive learning progress: the positive part of a fast-versus-slow loss estimate. The synthetic sampler normalizes and caps that score before softmax sampling, preventing an unbounded raw-loss tail.

For a differentiable richness functional $\mathcal{R}(U)$, richness-aligned surprise is

$$a_i = -\langle \nabla_U \mathcal{R}(U), \nabla_U \ell_i(U, C) \rangle, \quad \mathcal{R}(U - \eta \nabla \ell_i) - \mathcal{R}(U) = \eta a_i + O(\eta^2). \quad (2)$$

Positive a_i predicts a local increase in $\mathcal{R}$. Probabilities are mixed with uniform sampling and monitored by $\text{ESS} = 1 / \sum_i p_i^2$. We test log-determinant covariance, a covariance barrier, participation ratio, and a predictive-spectral functional. Coordinate weighting is kept distinct: it reweights latent coordinates inside Eq. (1), whereas ras scores observations by gradient alignment.

For factorization, each relation r (crop, color, blur, flip, or a time displacement) induces a fitted affine action $z' \approx A_r z + b_r$ in a whitened latent space. Given a shared orthogonal basis B partitioned into K blocks, F_K is the fraction of centered operator energy contained within those blocks. The key statistic is

$$\Delta F_K = F_K(\{A_r\}) - \mathbb{E}_{Q_r \sim \text{Haar}} F_K(\{Q_r^\top A_r Q_r\}), \quad (3)$$

where each independent conjugation preserves the spectrum, rank, and norm of its operator but destroys shared orientation. Thus $\Delta F_K > 0$ tests joint organization rather than marginal operator simplicity.

Table 1. Hypotheses and decisive checks. “Mixed” means that a weaker statement survived while the original general claim did not.

ID	Hypothesis	Status after experiments
H0	Stop-gradient JEPA with EMA has locally stable fast-slow dynamics near a non-isolated equilibrium manifold.	Supported locally in continuous time; neural-scale scope open.
H1	Persistence or transformations induce separable entity/relation subspaces.	Mixed: vector yes; pixels partial; operator blocks reproducible.
H2	A factorization score is a label-free substitute for downstream quality.	Rejected as universal; useful only as a within-run diagnostic.
H3	Learnable surprise prioritizes hard learnable events while avoiding noise.	Mechanism works in a toy mixture; not yet a robust video gain.
H4	Richness-aligned surprise improves JEPA representations.	Rejected on completed matched runs.
H5	Early, controlled factorization pressure improves later utility.	Not yet tested end-to-end; implementation smoke-tested.

§4. Experimental Methodology

4.1. Data, models, and metrics. The experimental ladder isolates increasingly realistic confounds: a vector world with a slowly switching entity and AR(1) context; Shapes3D, with shape as entity and five context factors [5]; Tiny ImageNet with official-style ViT-S/4 I-JEPA; and 20-frame Moving-MNIST,

where ten causal context frames predict ten future frames [8]. Static Shapes3D uses 16k samples and a patch CNN; the spatial variant uses multi-block masking and a 16-dimensional latent. Tiny ImageNet uses EMA target features and frozen ridge/MLP probes.

4.2. Metrics and controls. Utility uses held-out probes: entity/context classification, Tiny ImageNet accuracy, and low-shot velocity RMSE, FDE, and direction accuracy. We also record latent rank, corrupted draws, and ESS. The label-assisted counterfactual score is

$$\begin{aligned} d_{\text{diff}} &= \mathbb{E}[\|P_E z - P_E z'\|^2 \mid E \neq E', C = C'] , \\ d_{\text{same}} &= \mathbb{E}[\|P_E z - P_E z'\|^2 \mid E = E', C \neq C'] , \\ Q_E &= d_{\text{diff}} / (d_{\text{same}} + \epsilon). \end{aligned} \quad (4)$$

with P_E fitted only on training data and evaluated on held-out pairs. We report whitened values because the raw ratio can be inflated by scale.

The official I-JEPa audit fixes 10,000 test images and epochs 50–300. It uses held-out LDA/crop scores and flip, fixed-color, and blur operators with test $R^2 \geq .5$. We subtract $(\text{tr } A/d)I$, test $K = 2, 4, 8$, repeat at $d = 32, 56, 96$ with 100 bootstrap and Haar draws, and compare projectors fit on disjoint halves.

4.3. Reliability and research-process audit. The vector experiment compares temporal, shuffled, oracle, and random encoders across three seeds; all samplers have matched uniform baselines. Proxy tests include checkpoint differences, cross-method calibration, and future holdout. Earlier conclusions changed after finding a wide-bottleneck confound, cyclic-factor R^2 , wrong entity rank, gradient leakage, a scalar-identity operator component, and variable whitening dimension. All were corrected before claims were retained (Appendix C).

§5. Results

5.1. Surprise separates difficulty from learnability, but not reliably. In an easy/hard/noise synthetic mixture, raw MSE ordered examples as easy < hard < noise throughout training. It therefore could not distinguish informative challenge

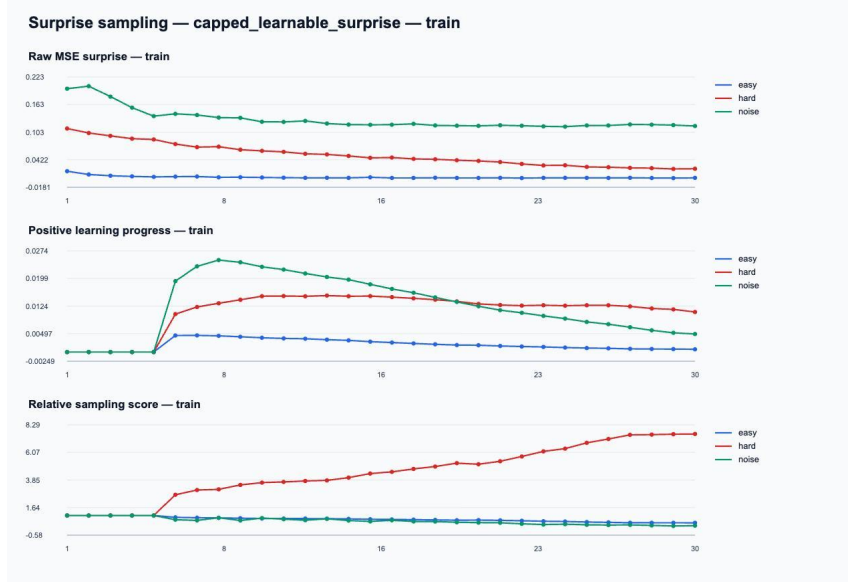


Fig. 1. Synthetic sampler sanity check extracted from the experiment log. Raw error ranks noise highest (top), whereas positive learning progress (middle) and the capped relative score (bottom) increasingly prefer hard, learnable examples.

from irreducible error. Positive learning progress and a capped relative score changed the batch composition: the hard fraction rose while the noise fraction stayed low (Fig. 1). This validates the intended mechanism, not a downstream benefit.

Moving-MNIST exposed the limits. After correcting positional encoding and pooling, learning rate 5×10^{-4} collapsed token rank to 15–25 and direction accuracy to .57–.59; 2.5×10^{-4} retained rank 42–51 and accuracy .73–.76. Background prediction loss was near zero while foreground loss was over an order of magnitude larger, so average surprise rewarded a shortcut. The adaptive sampler changed the distribution, increased noise share, and did not improve the frozen probe. Stable pretraining still helped low-shot velocity: RMSE improved from 4.03 to 3.49 at 128 labels and from 2.40 to 2.21 at 2048 labels (Appendix Fig. 8).

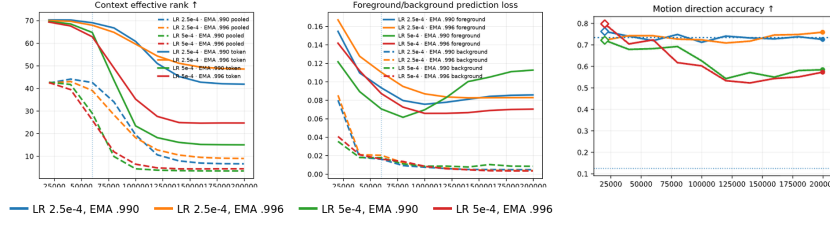


Fig. 2. Moving-MNIST stability audit. The higher learning rate causes dimensional collapse and weaker direction probes despite low background-dominated prediction loss. Diamonds mark early peak direction accuracy; circles mark the final checkpoint.

Table 2. Final matched sampling results. No learned sampler exceeds uniform on semantic accuracy.

Data	Method	Top/entity	Context/Top-5	Eff. rank	Loss/ESS
Shapes3D clean	Uniform	.852	.408	6.53	.073
	Loss	.817	.358	5.70	.094
	RAS logdet	.696	.510	8.14	.095
	RAS PR	.700	.428	6.69	.099
Shapes3D 20% noise	Uniform	.751	20% corrupt	–	16k ESS
	Loss	.713	15% corrupt	–	10-10 ² ESS
	RAS PR	.686	1% corrupt	–	1-10 ESS
	Clean oracle	.800	0% corrupt	–	12.8k ESS
Tiny IN ViT-S/4	Uniform cont.	24.7%	50.6%	44.08	.1378
	RAS cont.	23.2%	50.3%	43.64	.1367

5.2. Richness-aligned surprise detects corruption but loses support. Table 2 summarizes the matched runs. On clean Shapes3D, logdet ras raised effective rank to 8.14 but reduced entity accuracy from .852 to .696: spectral volume contained context, not better shape semantics. With 20% corruption, participation-ratio ras drew only 1% corrupted samples, but ESS fell to 1-10 and accuracy to .686; the support-preserving clean oracle reached .800. Rejection and coverage are distinct requirements.

On Tiny ImageNet, both methods continued from the same epoch-700 ViT-S/4 checkpoint. Predictive-spectral ras maintained ESS 39k-55k, improved loss from .1378 to .1367, but reduced Top-1 from 24.7% to 23.2%. Thus H4 is rejected on completed runs even without catastrophic support collapse.

5.3. Entity/context factorization is partial and capacity-dependent. In the three-seed vector world, whitened Q_E was .67-.95 for random encoders, .23-.97 after time shuffling, and 6740-8220 after temporal training, isolating persistence. Pixels were harder: temporal Shapes3D failed at dimensions 8/32, and nonlinear/CNN models collapsed under plain MSE+EMA. A stable spatial I-JEPA CNN increased shape accuracy from .421 to .917. A 3D subspace retained .915 shape accuracy while context fell to .127, versus .917/.414 in the full 16D latent. The complement still decoded shape at .786, so factorization is partial rather than $z_E \oplus z_C$.

5.4. Official I-JEPA: good monitors are not universal selectors. Within official I-JEPA epochs 50-250, exact LDA correlated with ridge/MLP/finetuned Top-1 at .976/.980/.961; crop stability correlated at .959/.968/.976. Online LDA matched exact LDA at Pearson .972. Selection tests overturned the optimistic reading: epoch-controlled crop correlation across training methods was about -.51; early-to-late ras crop forecasting had $R^2 = .014$; direct official-versus-ras ordering was right only 3/5 times; and online-LDA future R^2 was mostly non-positive (Appendix A). H2 fails in its strong form.

5.5. Joint block structure is real, early, and not monotone with quality. The corrected spectral-null test found $\Delta F_K > 0$ at every trained official checkpoint. At fixed $d = 32, 56, 96$, structure is high in epochs 50-100, falls by 150, and stays lower through 300; overlapping intervals do not locate an exact peak. For $K = 4, 8$, split-half projectors beat their own Haar null in all 18 dimension/epoch combinations ($Z = 10.5-57.8$), whereas $K = 2$ has near-random cases. The result is reproducible fine-block organization, not a universal binary split.

Downstream accuracy rises through epoch 250 while block structure declines after the early window (Fig. 3). This permits

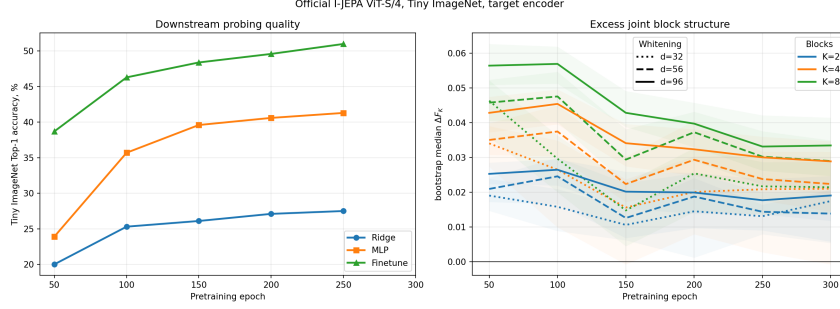


Fig. 3. Official I-JEPa on Tiny ImageNet. Ridge, MLP, and finetuning probes improve (left), while fixed-dimension excess joint block structure is highest early and lower from epoch 150 onward (right). Shading denotes bootstrap intervals.

a transient organization/consolidation interpretation, but rules out maximizing ΔF_K as a scalar quality objective.

In a common epoch-50 frame, affine predictability Q_{17} rose .650→.795, block effective rank fell .993→.761, and inter-block MI rose .006→.083. These five-point trends are phase diagnostics confounded by time, not validated objectives.

§6. Discussion

There is no context-free richness. The evidence separates marginal capacity (effective rank), relation-discriminative capacity (LDA), affine predictability (Q_{17}), modularity (ΔF_K), support/stability (ESS and split halves), and utility (probes). Improving one can harm another: logdet ras raises rank but lowers entity accuracy; predictive ras lowers JEPa loss but lowers Top-1; block structure declines as probes improve.

Factorization is a direction, not a finished score. The compact Shapes3D subspace and reproducible I-JEPa operator blocks are structural evidence; failed transfer and forecasting are evaluative counterevidence. We therefore pose, rather than claim, *factorize-compose-consolidate*: early relation-sensitive modules may later mix or compress into useful features. Only phase-specific interventions can test this mechanism.

Surprise needs a target and a support constraint. Raw loss asks what is unexpected, not learnable. Learning progress handles the toy distinction, but background dominates Moving-MNIST. ras inherits its target functional’s blind spots and can sacrifice coverage. Video sampling therefore needs local motion scores, capped influence, uniform mixing, an ESS floor, and a measured favorable post-update effect.

§7. Future Work Plan

Four gates follow: multi-seed fixed-d/cross-fitted measurement; heterogeneous Moving-MNIST then small V-JEPA profiling; early/constant/late/shuffled causal factorization; and only then motion-weighted surprise with uniform mixing and an ESS floor. Residual- Q_{17} passes tests, resume, official-I-JEPA steps, and a 22.6-GB V100 smoke, but full training is unrun. Success is $\geq 3\%$ velocity-RMSE or .3-pixel FDE improvement without direction loss across seeds (Appendix B).

§8. Contribution

Andrey Vorfolomeev	Factorization framing; fast-slow EMA theory; hypotheses; synthesis/reporting.
Konstantin Kostikov	RAS; I-JEPA infrastructure; sampling/transfer/null audits; residual- Q_{17} ; reproducibility.
Alexander Korolev	Surprise samplers; Moving-MNIST; stability/shortcut audits; motion probes/objective.
Mikhail Davydov	Shapes3D Q metrics; correlations/timing; bandit analysis; shared artifacts.

Mark Obozov served as curator/supervisor. This export-derived statement requires author confirmation; reported claims were rechecked against archived artifacts.

§9. Conclusion

No learned sampler beat uniform, and neither factorization nor surprise is a universal quality scalar. The evidence instead supports relation- and phase-specific structure. Cross-fitting, causal schedules, collapse checks, and support constraints define the resulting video-representation program; limitations are detailed in Appendix E.

References

1. Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas, *Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture*, Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 15619–15629.
2. Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mahmoud Assran, and Nicolas Ballas, *Revisiting Feature Prediction for Learning Visual Representations from Video*, arXiv:2404.08471, 2024.
3. Adrien Bardes, Jean Ponce, and Yann LeCun, *VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning*, International Conference on Learning Representations, 2022.
4. Quentin Garrido, Randall Balestriero, Laurent Najman, and Yann LeCun, *RankMe: Assessing the Downstream Performance of Pretrained Self-Supervised Representations by Their Rank*, Proceedings of the 40th International Conference on Machine Learning, PMLR 202, 2023, pp. 10929–10974.
5. Hyunjik Kim and Andriy Mnih, *Disentangling by Factorising*, Proceedings of the 35th International Conference on Machine Learning, PMLR 80, 2018, pp. 2649–2658.
6. Etai Littwin, Omid Saremi, Madhu Advani, Vimal Thilak, Preetum Nakkiran, Chen Huang, and Joshua Susskind, *How JEPА Avoids Noisy Features: The Implicit Bias of Deep Linear Self-Distillation Networks*, Advances in Neural Information Processing Systems 37, 2024.
7. Sören Mindermann, Jan Brauner, Muhammed Razzak, Mrinank Sharma, Andreas Kirsch, Winnie Xu, Benedikt Höltingen, Aidan Gomez, Adrien Morisot, Sebastian Farquhar, and Yarin Gal, *Prioritized Training on Points that are Learnable, Worth Learning, and Not Yet Learnt*, Proceedings of the 39th International Conference on Machine Learning, PMLR 162, 2022, pp. 15630–15649.
8. Nitish Srivastava, Elman Mansimov, and Ruslan Salakhutdinov, *Unsupervised Learning of Video Representations Using LSTMs*, Proceedings of the 32nd International Conference on Machine Learning, PMLR 37, 2015, pp. 843–852.
9. Vimal Thilak, Chen Huang, Omid Saremi, Laurent Dinh, Hanlin Goh, Preetum Nakkiran, Joshua Susskind, and Etai Littwin, *LiDAR: Sensing Linear Probing Performance in Joint Embedding SSL Architectures*, International Conference on Learning Representations, 2024.

Сведения на русском языке

**А. Ворфоломеев, К. Костиков, А. Королев,
М. Давыдов.**

Факторизация и неожиданность как сигналы управления обучением joint-embedding predictive architectures.

Аннотация. В работе исследуются два возможных сигнала без разметки для обучения и диагностики JERA: неожиданность наблюдения и факторизация латентного пространства. Серия контролируемых экспериментов проведена на синтетической векторной модели, Shapes3D, Tiny ImageNet и Moving-MNIST. Richness-aligned surprise обнаруживает зашумлённые примеры, но чрезмерно концентрирует распределение выборки и не превосходит равномерный базовый метод. В латентных представлениях обнаруживается частичная факторизация: компактное подпространство Shapes3D сохраняет информацию о форме, а операторы преобразований официальной I-JERA имеют воспроизводимую совместную блочную структуру сверх спектрального нулевого контроля. При этом блочная структура максимальна на ранней стадии и затем уменьшается, хотя качество downstream-зондов растёт. LDA- и стор-метрики хорошо отслеживают одну траекторию обучения, но ненадёжно сравнивают разные методы и прогнозируют будущее качество. Поэтому предложен многомерный relation-conditioned профиль качества и план причинных экспериментов, объединяющий раннюю затухающую регуляризацию факторизации, surprise-сэмплирование, контроль коллапса и ограничение эффективной поддержки выборки.

Ключевые слова: JERA; обучение без учителя; факторизация представлений; surprise-сэмплирование; диагностика представлений.

Appendix §A. Extended Experimental Tables

Table 3. Official I-JEPA temporal factorization, $K = 4$. Probe values are unavailable at epoch 300 in the archived table.

Epoch	Ridge	MLP	Tune	Q10 erank	Q11 MI	Q17 R^2	CKA ₅₀
50	.200	.239	.387	.993	.006	.650	1.000
100	.253	.357	.463	.793	.060	.718	.783
150	.261	.396	.484	.781	.075	.781	.708
200	.271	.406	.496	.764	.078	.787	.655
250	.275	.413	.510	.774	.081	.793	.634
300	–	–	–	.761	.083	.795	.632

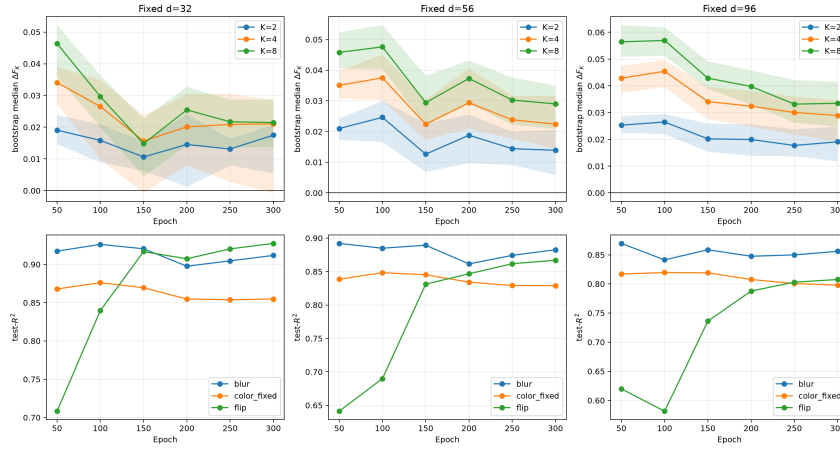


Fig. 4. Fixed-dimension robustness audit. Top: bootstrap distributions of ΔF_K . Bottom: held-out operator fit. The high early window and post-150 decline persist at $d = 32, 56, 96$.

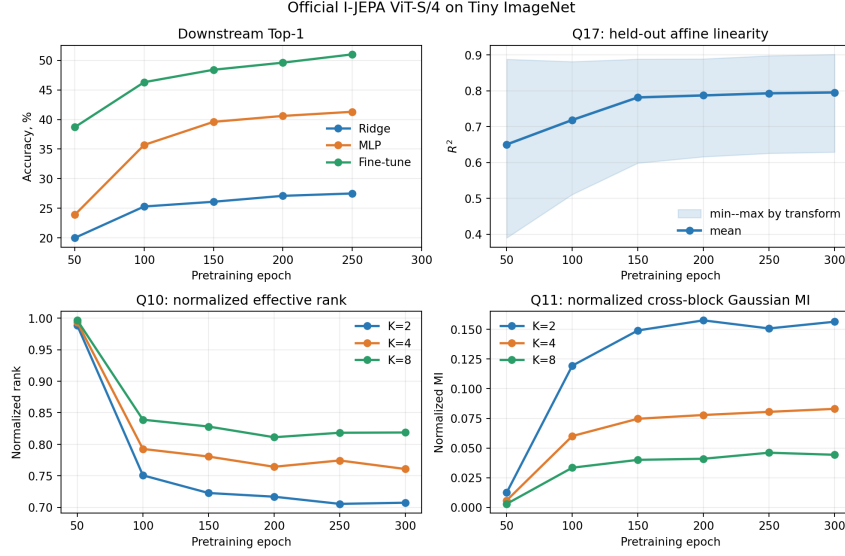


Fig. 5. Temporal factorization trajectories for official I-JEPA. $Q17$ predictability rises, block capacity contracts, and inter-block dependence increases. The common time trend prevents a causal interpretation of the correlations.

Appendix §B. Future-Work Decision Matrix

Appendix §C. Metric and Implementation Audit

Appendix §D. Local Fast-Slow Result: Scope and Assumptions

We study the deterministic system

$$\dot{U} = -F(U, C), \quad \dot{C} = \varepsilon(PU - C), \quad 0 < \varepsilon \ll 1,$$

Shapes3D with 20% Gaussian noise: can sampling reject bad data?

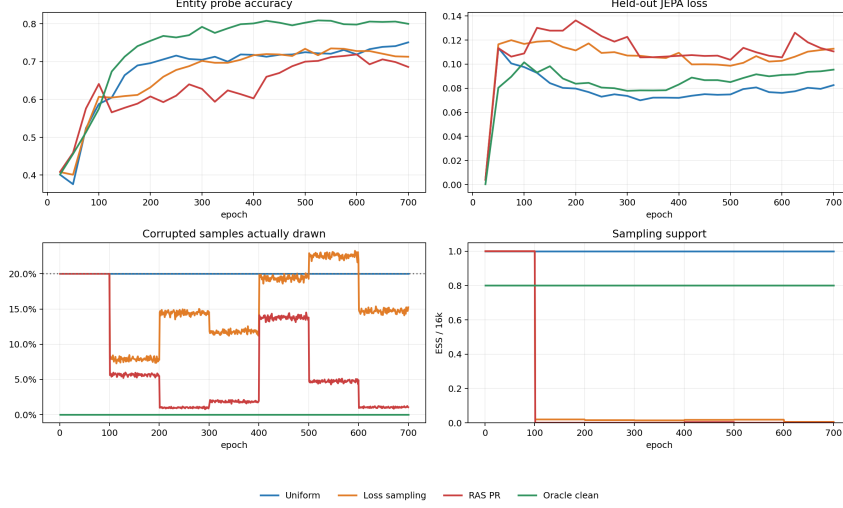


Fig. 6. Corrupted Shapes3D diagnostic. RAS rejects Gaussian noise but collapses sampling support and underperforms both uniform sampling and the support-preserving clean oracle.

Here F is the online gradient of the stop-gradient objective plus a regularizer. The fast subsystem may have a non-isolated critical manifold. Under local smoothness, normal attraction, stable fibers, and exponential attraction of the reduced slow residual, a nearby invariant slow manifold persists and Eq. (5) follows. A nonlinear tanh example with $O(2)$ symmetry verifies non-vacuous assumptions. The result is neither global nor an SGD theorem; here it motivates phase-specific, state-dependent interventions.

$$\text{dist}((U(t), C(t)), \mathcal{E}_1) \leq M(e^{-\gamma_f t} + e^{-\gamma_s \varepsilon t}) \text{dist}((U(0), C(0)), \mathcal{E}_1). \quad (5)$$

Appendix §E. Limitations

Most image results use one seed and five or six checkpoints. Spatial Shapes3D factorization is label-assisted; Tiny ImageNet

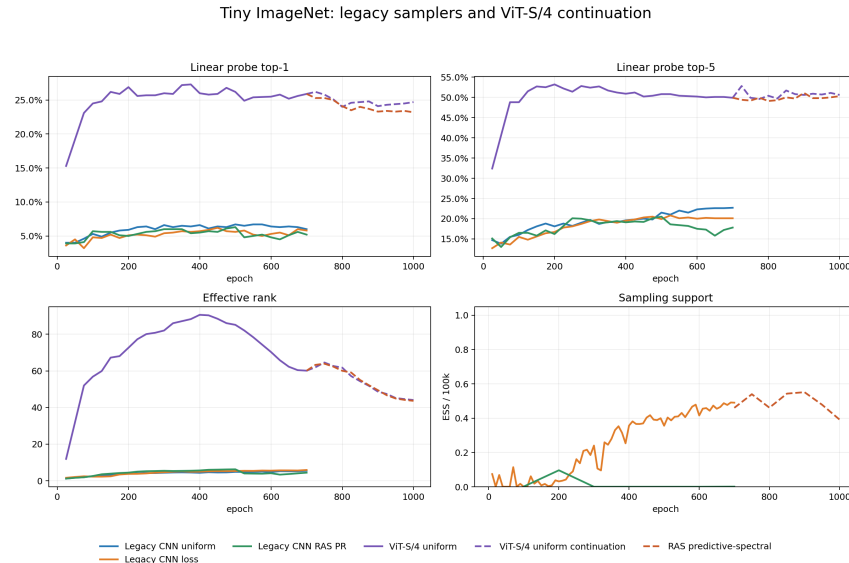


Fig. 7. Tiny ImageNet sampling trajectories. Moderate-support predictive RAS improves held-out JEPA loss slightly but not semantic probing.

is small; and one-digit Moving-MNIST is homogeneous. Training time confounds many correlations. The local theorem is deterministic and continuous-time. Residual- Q_{17} and the video interventions have smoke tests but no completed causal run. These limits prohibit claims of a universal metric, an established phase transition, or an improved sampler.

Appendix §F. Reproducibility Map

The supplied snapshot records code state, patches, manifests, compact outputs, TensorBoard events, configurations, and SHA-256 checksums. Datasets, 91.8 GiB of checkpoints, caches, and Git pack files were excluded; paths and sizes remain in the manifest. Principal entry points are:

- `scripts/analysis/evaluate_upstream_ijepa_spectral_block_robustness.py` for fixed-d spectral nulls;

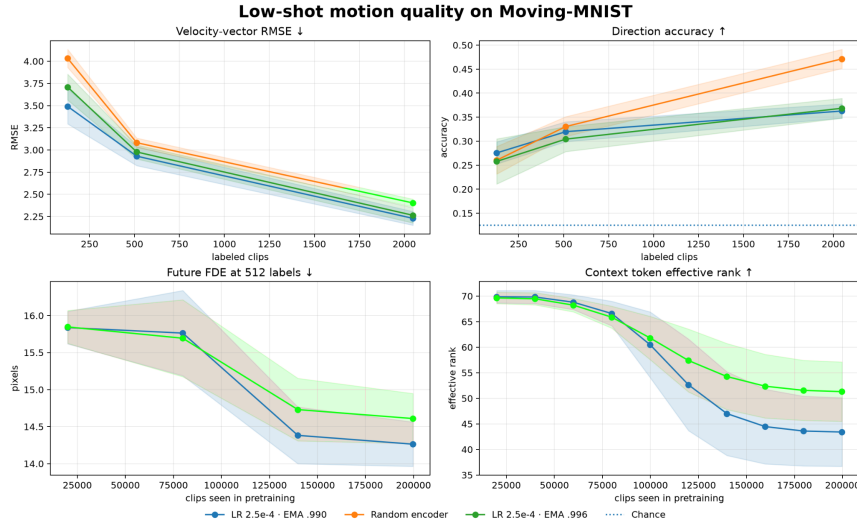


Fig. 8. Moving-MNIST low-shot probes. Stable pre-trained encoders improve velocity estimation over a random encoder; the gains do not validate the adaptive sampler.

Table 4. Precommitted gates for the next experimental cycle.

Gate	Minimum experiment	Pass condition	Stop/redirect condition
Measurement	≥ 3 seeds; fixed d ; cross-fitted blocks; leave-run-out calibration	Low checkpoint regret and stable split-half excess over Haar null	Same-run correlation without transfer or stability
Video profile	Two digits with crossings, speed bins, and bounces; then small V-JEPa	Relation metrics respond to motion, not background or epoch alone	Foreground/background audit shows shortcut dominance
Causal factorization	Early-decay, constant, late, shuffled, geometry-only	Early schedule improves probes across seeds without rank loss	Only auxiliary loss improves, or gain matches shuffled control
Surprise action	inter-Motion-local progress score; uniform mixture; ESS floor	$\geq 3\%$ velocity-RMSE or .3-pixel FDE gain, no direction loss	Support collapse, noise preference, or no matched uniform gain

- `scripts/analysis/evaluate_upstream_ijepa_temporal_factorization.py` for $Q5$ - $Q22$ trajectories;

Table 5. Corrections that changed or narrowed the interpretation.

Issue	Why it was misleading	Resolution and effect
Wide bottleneck	A random encoder preserved essentially all low-rank factors.	Reduced latent dimension; added random baselines and validation warning.
Cyclic-factor R^2	Hue values near 0 and 1 are neighbors but linear regression treats them as distant.	Replaced with classification per discrete factor.
Wrong entity k	Four centered shape means have rank three, not two.	Set $k = C - 1$; whitened Q_E changed from 5.87 to 2.47.
Scale-sensitive Q_E	Covariance growth inflated the raw ratio.	Whitened features and reported covariance/rank first.
Gradient leak	A denominator allowed the model to reduce normalized loss by inflating variance.	Detached normalization statistics.
Aggregate mislabel	A reported “full” accuracy was read from the entity-subspace field.	Corrected field mapping and regenerated reports.
RNG desynchronization	Entity switches changed later context random numbers.	Split random streams for entity and context.
Scalar identity in F_K	cI is block diagonal in every basis, including untrained/random controls.	Centered $A \leftarrow A - (\text{tr } A/d)I$ and removed invalid grayscale.
Variable whitening d	Apparent epoch-100 maximum could reflect $d = 8 \rightarrow 152$.	Repeated at fixed $d = 32, 56, 96$ with 100 bootstrap/null draws.
Same-run correlation	Time made many metrics look predictive.	Added differences, future holdout, cross-method transfer, and checkpoint-regret plan.

- `scripts/analysis/evaluate_factorization_proxy_transfer.py` and `evaluate_online_lda_replay.py` for transfer and forecast audits;
- `scripts/analysis/audit_ras_local_utility.py` and `image-training launchers` for sampling tests;
- `third_party/ijepa_q17/` and the `residual-Q17 launcher/configuration` for the not-yet-run causal pilot.

For a reproducible paper release, the next step is to publish a sanitized repository revision with environment lockfiles, the compact output bundle, exact checkpoint identifiers, and a single command that regenerates every table and figure. Any credentials or private infrastructure identifiers from communication exports must remain excluded.

Affiliations and corresponding-author metadata to be confirmed before submission.